Severe Acute Malnutrition Clinical case presentation

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MVJ MC & RH

Patient particulars

Name xyz

Age 9 months

Sex female

Address: Hoskote

Religion: Hindu

Informant mother reliable

Date of admission 10th may 2020 7am

Date of examination 10th may 2020 9 am

Chief Complaints

- Diarrhoea since 3 days
- Irritability since 1 day

History of presenting illness

The child was apparently normal 3 days back when child started vomiting , the vomitus contained breast milk following which the child had loose stools which was sudden in onset ,clear with 2-3 episodes on the first day less in quantity and increased to 6-7 episodes on the third day and was about 50 ml in total and the child tends to cry when passing stools.

There is no H/O of blood or mucous in stools, or foul smelling stools

The child was initially playfull on the first day but has become more irritable and cries more frequently since two days, the child also refuses any complementary feed and only takes breast milk intermittently.

The child is also thirsty and the frequency of urination also has decreased to only one time yesterday.

.

There is no H/O

- Fever
- Distention of abdomen
- Convulsions

Child is not bottle fed and has not been given any food from outside

Past History

- Similar episode of diarrhoea one month back that was self limiting
- No history of vaccine preventable diseases in the past
- Child has decreased appetite since the last episode of diarrhoea on month back

Treatment history

- The mother did not give any treatment and brought the child to hospital on worsening of the symptoms

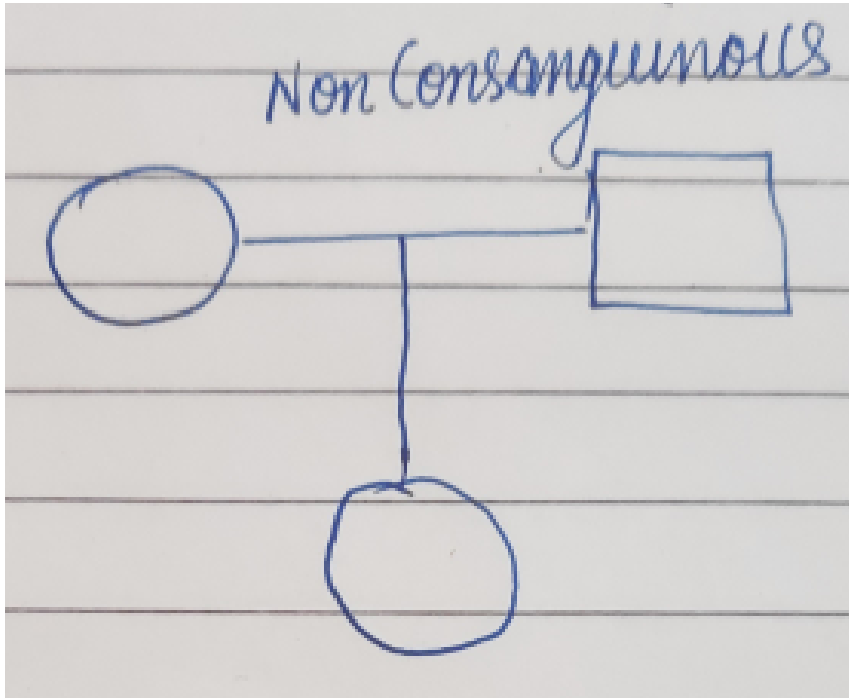
Family history

No significant family history

No history of recent travel

No similar complaints of diarrhoea in the family

No history of TB in any of the family members



Birth history

Pregnancy

Booked at MVJ MC & RH

Pregnancy confirmed after 2 months of ammenorhea

Total of 6 antenatal visits

Total weight gained 12.5 kg

Period of gestation: 38 weeks

Antenatal History

1ST Trimester

Folic acid Taken

Dating scan done

No H/O of UTI, fever, rash, lymphadenopathy

exposure to radiation or drugs

No H/O of hypothyroidism

2ND *Trimester*

Quickening at 20 weeks

2 doses of TT taken 1 month apart

Iron, folate and calcium taken

Anomaly scan done, OGTT done, NT scan done all are normal

No H/O headache, blurring of vision, pedal edema

No H/O polyuria, polydipsia

No H/O fever with rash, lymphadenopathy

No H/O bleeding PV

3RD *Trimester*

Appreciated fetal movements well

Growth scan done

No H/O preeclampsia, headache, blurring of vision, pedal edema

No H/O bleeding PV, leaking PV, foul smelling discharge

No H/O maternal infections

No H/O PROM

Labour

Time and date : 30 August 2019 ,5pm

Full term normal vaginal delivery

Postnatal history

Birth Weight : 2.8 Kg

Baby cried immediately after birth

No NICU admission and no post natal complications

Breast feeding was initiated ½ an hour after delivery

Given adequately on demand during day and night

DOMAINS	MILESTONE	AGE OF ATTAINMENT	EXPECTED AGE	DQ
GROSS MOTOR	Neck Holding	3 month	3 month	
	Rollover	6 month	6 month	
	Sit with support	6 month	6 month	
	Sitting without support & crawling	8 month	8 month	
FINE MOTOR	Hand regard	3 month	3 month	
	Bidextrous grasp	4 month	4 month	
	Unidextrous grasp & transfer of objects	8 month	8 month	
SOCIAL	Social smile	2 month	2 month	
	Mother regard	3 month	3 month	
	Mirror play	6 month	6 month	

LANGUAGE	Vocalise Cooing Monosyllable	2 month 3 month 6 month	2 month 3 month 6 month	
Visual fixation		4 weeks	4 weeks	

Immunisation

- At birth BCG, OPV, Hep B1
- 6 weeks DTwP1, Hep B2, Hib1, PCV1
- 10 weeks DTwP2, Hib2
- 14 weeks DTwP3, Hib3, PCV3
- 6 months OPV1, Hep B3

Nutrition

- Exclusively breast fed for the till 7th month
- Complimentary feeding was intiated at the 8 th month with 1-2 breast feeds per day
- No history of allergy to any food .
- Based on the 24 hour recall method
- Morning : 1 idly and two to three teaspoon sambar = 150kcal + 1 gm protein
- Afternoon : 2 chapathi medium size and ¼ katori dal= 160 kcal + 6.25gm protein
- Breast feeding given only once at night and morning 100ml = 130kcal + 2.2 gm of protein
- Total : 440 kcal + 9.47gm of protein
- Required calorie intake 830 Kcal and protein intake 14.11 gm
- Deficiency of 390 kcal and 4.64 gm of protein.

Socio economic status

- Lives in a pucca house
- Belongs to class 4 socioeconomic status (modified BG Prasad classification)
- Source of water is borewell
- There is no open defecation around the house

Summary

- 9 month old girl of birth order one ,who was born of non consanguineous marriage via normal full term normal vaginal delivery was brought to the OPD with history of clear watery diarrhoea with increasing episodes since 3 days maximum of 6 episodes on 3rd day with increased irritability , thirst and decreased appetite since two days and not gaining weight since 2 months started on complementary feed of 440kcal and 9.47gm of protein per day and breast fed twice a day which is inadequate for a child of 9month
- With this I consider that the child is suffering from acute gastroenteritis with some dehydration and malnutrition

Examination

- Child is on the mother lap and crying intermittently,
- Child was conscious but also irritable

Vitals

- Temperature : afebrile 95F
- Airway- clear
- Respiratory rate- 45 cycles per minute , predominantly abdomino-thoracic , pattern and depth normal , no usage of accessory muscles of respiration
- sPO2 : 98%

- Pulse – 140 bpm normal in character ,regular in rhythm , low volume pulse with no radioradial or radiofemoral delay.
- Blood pressure-60/50 mm hg lower side of normal
- Capillary refilling time 2sec

ANTHROPOMETRY

	Actual	Expected	Inference
Weight	6.5 kg	8.2 kg	< -3 S.D
length	69 cm	70.1 cm	< - 2 S.D
Weight for length	6.5 kg	8 kg	< -3 S.D
Head circumference	43.5 cm	44cm	50th centile
Chest circumference	40cm	41cm	50th centile
Mid arm circumference	11cm	13.5cm	< -3 S.D

- Head to toe examination

Head: normal in size and shape, sunken anterior fontanelle,

Hair: easily plucable.

Eyebrows: normal ,

Face:, wrinkled appearance of face

Eyes: Sunken appearance of eyes, no signs of vit A deficiency

Ears: normal

Oral cavity: dry , no cleft lip or palate, tongue is dry, no dentition, no signs of bcomplex deficiency.

Neck: prominence of muscles of the neck

Chest: movement of the chest is seen, rib markings are prominent as subcutaneous fat is lost

Umbilicus: normal

Abdomen: no distension

Genitalia: normal

Back: prominence of scapula , decreased subcutaneous fat over the buttocks

Extremity: normal arms and legs show muscle wasting,

Skin: dry, loss of subcutaneous fat, skin pinch test 2.5 seconds

Feet: normal , no edema

Orifices: normal

Gastro intestinal system
Inspection
Shape of abdomen : normal
All quadrants move equally with respiration
Umbilicus central
No sinuses/scasrs/dilated veins
No visible peristalsis
Undue prominence of both the anterior superior and posterior superior iliac spines
Skin is dry and shiny
Hernial orifices normal
Palpation
Baby cries on pressing the periumbilical region, no organomegally, no free fluid
Percussion normal
Bowel sounds normal

Systemic examination

- CNS

GCS 15

Higher mental functions

Conscious, irritable, responds to oral command

Cranial nerve examination

Cranial nerve	Examination	Right	left
optic	Recognises mother	Present	Present
	Puppilary reflexes	Red reflex present	Red reflex present
Occulomotor nerve	Able to follow objects in all directions of gaze	Present	Present
	Light reflex direct & indirect	Present	present
Trigeminal nerve	Corneal reflex	present	present

Facial nerve	**Eamination**	**normal**	**normal**
Vestibulococchlear nerve	Responds to sound	present	normal

Motor system	Upper limb		Lower limb	
	right	left	right	left
bulk	Wasting of muscles seen symetrically		Wasting of muscles seen symetrically	
Tone	hypotonia	hypotonia	hypotonia	hypotonia
inspection	normal	normal	normal	normal
palpation	normal	normal	normal	normal
Resistance to passive movement	Present	Present	present	present

Power

Upper limb	3+	3+
Lower limb	3+	3+
neck	Able to lift head	
trunk	Able to turn around in bed	

Reflexes

Superficial reflex	
Plantar response	extensor
Deep tendon reflex	
Biceps reflex	normal
Supinator reflex	normal
Triceps reflex	normal
Knee jerk reflex	normal
Ankle jerk reflex	normal

Developmental assessment

No primitive reflexes seen , no abnormal movements

Sensory system

Pain- responded by crying

Other sensations could not be elicited

Respiratory system

Inspection

Shape normal

Undue prominence of ribs

Symetrical movement of both sides

Palpation : normal

Auscultation

Normal vesicular breath sounds heard

No added adventitial sounds

CVS

Inspection

JVP not elevated

Palpation

Apex beat felt

No parasternal heave

Auscultation

S1 S2 normal no added sounds

- Diagnosis

Acute gastroenteritis with some dehydration and severe acute malnutrition